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from airflow import DAG
from airflow.operators.python_operator import PythonOperator
from datetime import datetime
import pandas as pd
import os

default_args = {
    'owner': 'airflow',
    'start_date': datetime(2024, 1, 1),
    'retries': 1
}

dag = DAG(
    'us_credit_bias_audit_pipeline',
    default_args=default_args,
    description='DAG for ingesting and preprocessing US credit applicant data',
    schedule_interval='@daily',
    catchup=False
)

def ingest_data():
    url =
    "https://raw.githubusercontent.com/IN_hub-lap/CreditFairnessAudit/bias-audit-us-only/us_applicants_bias_audit_only.csv"
    df = pd.read_csv(url)
    os.makedirs("/tmp/us_credit_pipeline", exist_ok=True)
    df.to_csv("/tmp/us_credit_pipeline/raw_data.csv", index=False)

def preprocess_data():
    raw_path = "/tmp/us_credit_pipeline/raw_data.csv"
    df = pd.read_csv(raw_path)
    df.dropna(inplace=True)
    df['income_bracket'] = pd.cut(df['annual_income'], bins=[0, 30000, 60000, 100000, float("inf")],
    labels=['Low', 'Mid', 'High', 'Very High'])
    df.to_csv("/tmp/us_credit_pipeline/processed_data.csv", index=False)

t1 = PythonOperator(
    task_id='ingest_us_data',
    python_callable=ingest_data,
    dag=dag
)

t2 = PythonOperator(
    task_id='preprocess_us_data',

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python_callable=preprocess_data,  
dag=dag  
)
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t1 >> t2
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